
Aim: Study of interfacing using Mobile/web to publish or remotely access the data on the internet

Apparatus Required:

- Laptop / PC
- Smartphone
- Internet connection
- Blynk IoT platform (cloud-based)

Software Required:

- Blynk IoT Web Dashboard (<https://blynk.cloud>)
- Blynk Mobile Application

Theory:

The **Internet of Things (IoT)** is a system of interconnected devices that communicate and exchange data over the internet. These devices can be sensors, actuators, or controllers.

Blynk IoT platform provides a cloud-based environment that allows users to:

- Monitor data remotely
- Control devices using mobile/web interface
- Store and process data in real-time

Working Principle:

1. The user interacts with the **mobile/web dashboard**
2. Data is sent to the **Blynk Cloud Server**
3. The cloud processes and stores the data
4. Data is then accessed or displayed on another interface

In This Experiment:

- Virtual pins (V0, V1) act as communication channels
- Button widget → sends data
- Value widget → receives/displays data
- No physical hardware is required

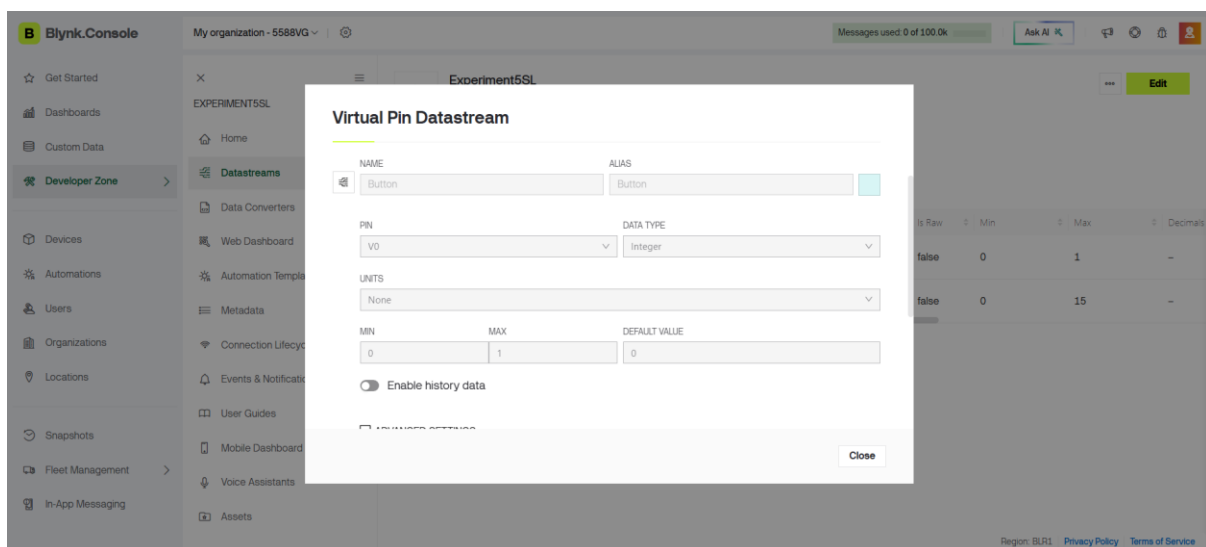
System Architecture Explanation:

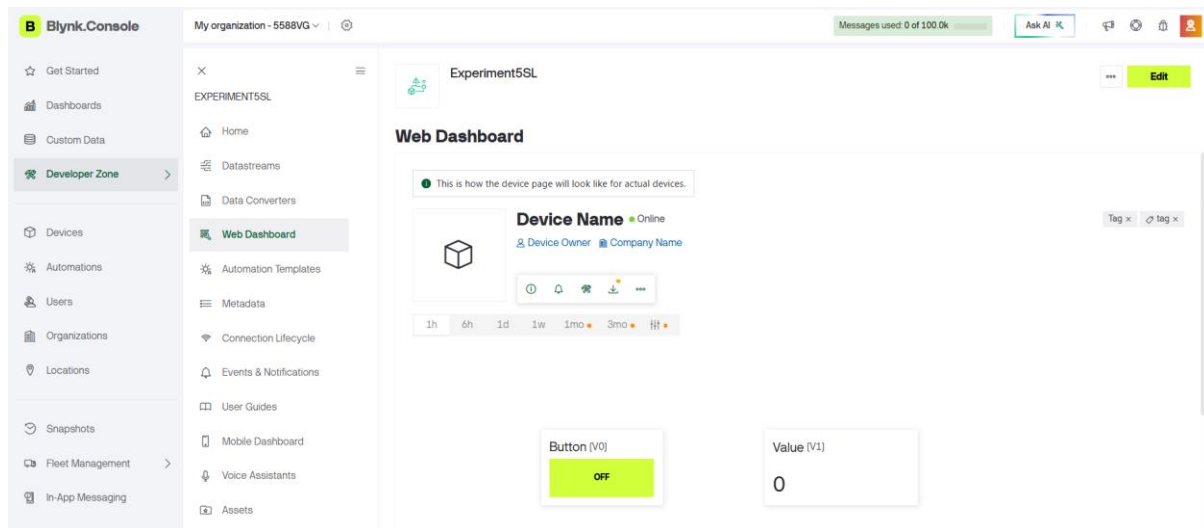
- **Frontend:** Mobile App / Web Dashboard
- **Backend:** Blynk Cloud Server
- **Communication:** Internet (WiFi)

- **Interface:** Virtual Pins

Procedure (Detailed):

1. Open the Blynk IoT platform using a web browser.
2. Create a new template and select:
 - Hardware: ESP8266
 - Connection: WiFi
3. Save the template.
4. Navigate to **Datastreams** section.
5. Create two virtual datastreams:
 - V0 → Integer (Button control)
 - V1 → Integer (Data display)
6. Open the **Web Dashboard**.
7. Add widgets:
 - Button widget → linked to V0
 - Value/Label widget → linked to V1
8. Configure button mode as **Switch**.
9. Save the dashboard.
10. Test the system by:
 - Toggling button (V0)
 - Manually updating value of V1
11. Observe real-time updates on the dashboard.



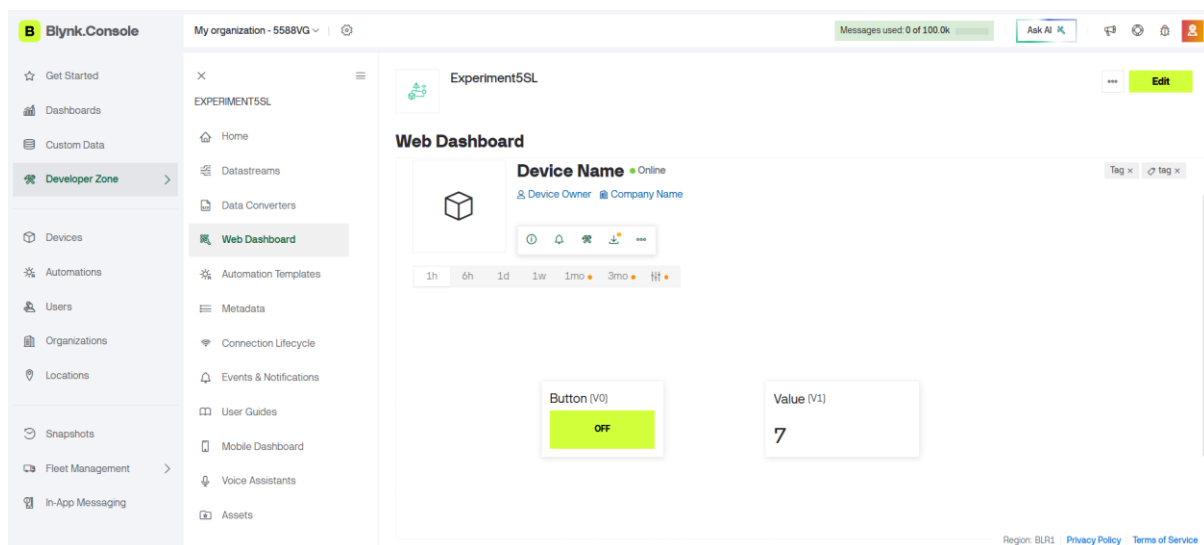


Observation (Detailed):

1. When the button widget is pressed, data is transmitted to the cloud through virtual pin V0.
2. The value widget displays the data received from virtual pin V1.
3. Data transmission occurs instantly via internet connection.
4. The web dashboard reflects changes in real-time.
5. Remote access to data is successfully achieved without physical hardware.
6. The system demonstrates bidirectional communication (send & receive).

Output:

- Button widget changes state (ON/OFF)
- Value widget displays updated values (0, 10, 50 etc.)
- Real-time synchronization between interface and cloud



Result

The experiment to study interfacing using mobile/web applications for publishing and remotely accessing data over the internet was successfully performed using the Blynk IoT platform.

Conclusion

Successfully studied interfacing using Mobile/web to publish or remotely access the data on the internet